



Understanding Transducers and Sensors

This presentation explores the fundamental concept of transducers, their different types, and their essential roles in various technological applications.



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Physical, Chemical Sensors and Biosensors

What is Transducer

A transducer is an electronic device which converts energy from one form to another.

Primary and Secondary Transducers

Primary Transducers

Directly convert physical parameters into electrical signals. Examples include pressure sensors, temperature sensors, and accelerometers.

Secondary Transducers

Convert an electrical signal from a primary transducer into a more useful form, like a digital output. Examples include signal amplifiers and converters.

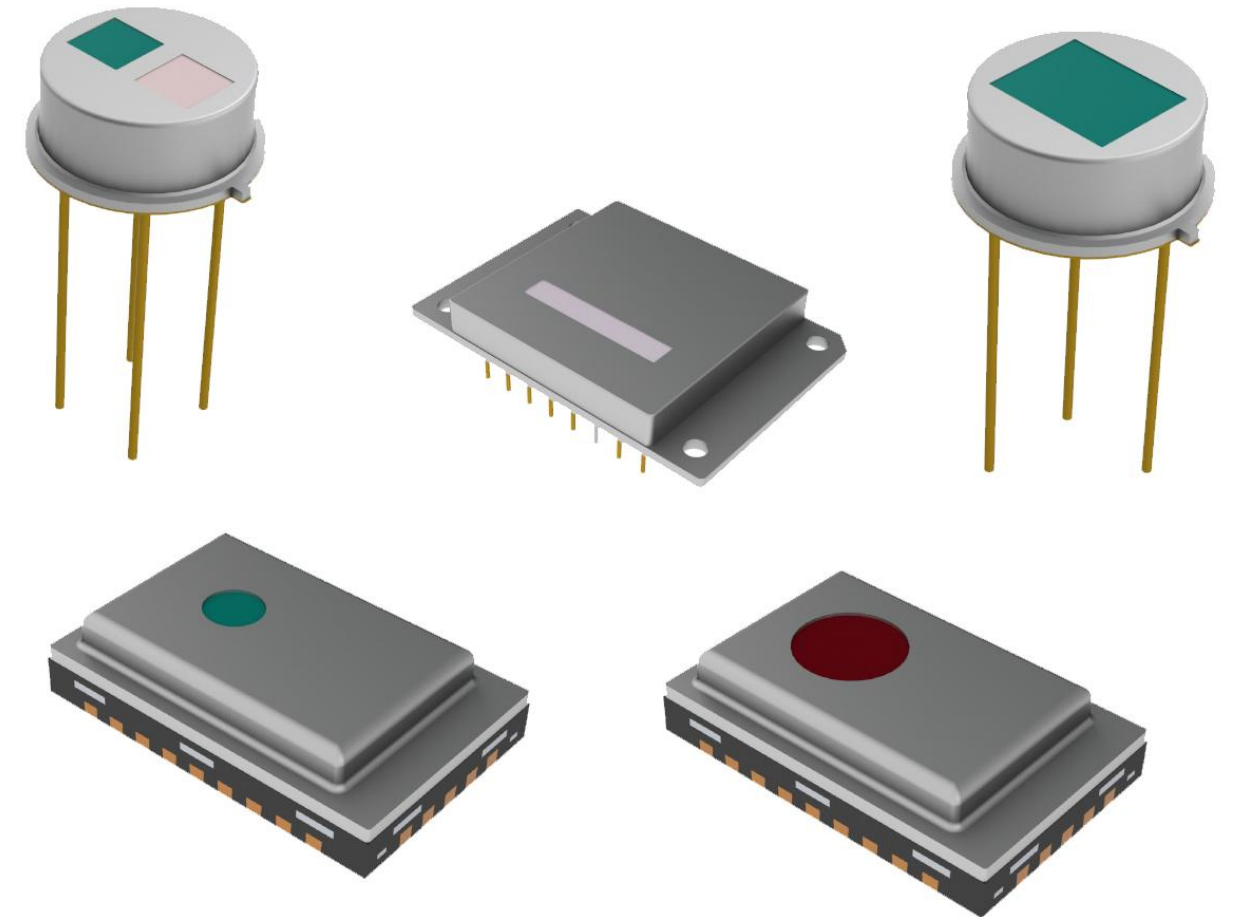
Active Transducers

➤ An active transducer is a **self-generating** type transducer that can produce an electrical voltage or current output without utilizing any external power source. These transducers draw the energy demanded for their operation from the measuring system

> This transducer **doesn't need an extra source**. Its design is not complicated, and the resolution is low. The best example is PV cell, Thermocouple, Tachogenerator, etc.

Passive Transducers

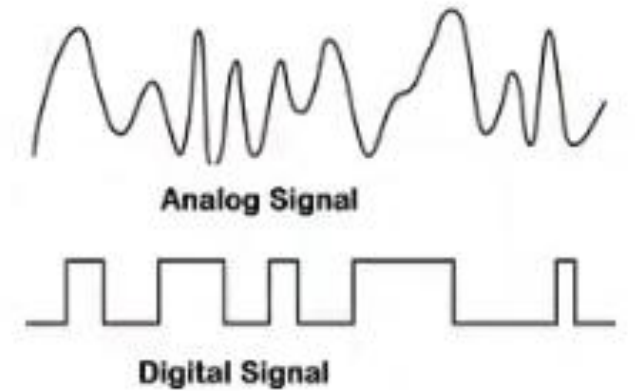
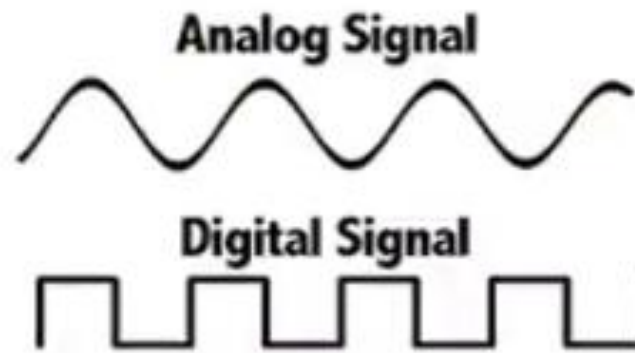
- The transducer **which needs an external power supply** for its operation is called Passive Transducer
- Some common examples of Passive transducer are LDR(Light Depending Resistor), Thermistor, LVDT(Linear Variable Differential Transformer), Potentiometer, Rheostat, Strain Gauge, etc.



Active Transducer	Passive Transducer
➤ Active transducer is self generating type transducer	➤ Passive transducer is called externally powered transducer
➤ Does not require power supply	➤ Requires external power supply
➤ Example. Thermocouple, Photovoltaic cell	➤ Example. RTD , Thermistor
➤ Design is simple	➤ Design is complicated
➤ Low Resolution	➤ High Resolution



Analog and Digital Transducers



Whether Output is **continuous** or in **discrete steps** the

transducers are classified as;
Analog and **Digital transducers**



Analog Transducer:

Converts Input quantity into an Analog output

[continuous function of time]

e.g. Strain Gauge, LVDT, Thermocouple and Thermister are some Analog Transducers.

Digital Transducer: Converts input quantity into Electrical o/p in the form of pulses represented as 0 or 1.

Eg; digital temperature sensor

